WASHINGTON STATE UNIVERSITY

Department of Mathematics & Statistics
College of Arts & Sciences

Bala Krishnamoorthy
Director of Graduate Programs, Professor
bala-krishnamoorthy.github.io

November 9, 2026

Search Committee, Tenure-Track or Tenured Position in Applied Mathematics
Department of Mathematics, University of Utah

Dear Colleagues,

I am applying to the faculty position in Applied Mathematics open in your Department. I am a Professor of Mathematics at Washington State University. A move to University of Utah will be a big step forward in my career, and offer me a professional challenge. At the same time, given my expansive and well-funded research program founded on geometry, topology, optimization, and their intersections, with applications in diverse domains including computational chemistry, additive manufacturing, electrical power grid, biomedical data analysis, phenomics, criminology, machine learning, and more, I am an ideal candidate for this position.

I have raised over $19.68M in research funding (total amount, as PI or co-PI). I have worked in large team projects involving experts from multiple disciplines. My funded projects have involved collaborators from medical schools, criminology, chemistry, agronomy, computer science, electrical engineering, and mathematics. On the other hand, I have been the sole PI of a research grant from NSF. I am confident of writing successful grant proposals with collaborators from Utah.

In fact, I already have long-term collaborations ongoing with Aurora Clark (Chemistry) and Bei Wang (Computer Science) from Utah. I have worked with both of them on medium and large-scale proposals as well as funded projects. My work in discrete optimization and its connections to other fields such as topology and geometry could compliment the expertise in your Department that forms the focus of your NSF RTG on Optimization and Inversion. Furthermore, my collaborative work with Aurora Clark will fit well with the material topology theme, and my work on additive manufacturing will fit nicely with the design emphasis theme, of this RTG. More broadly, I would enjoy collaborating with several of your faculty, including Andrej Cherkaev and Will Feldman on mathematical aspects of material sciences, Akil Narayan and Braxton Osting on optimization and scientific computing, and others. I also note that your have a different NSF RTG on Algebraic Geometry and Topology. My work on computational topology and applied geometric analysis could help build new bridges between the pure and applied research groups in your department. Apart from colleagues in Mathematics, I can foresee building collaborations with researchers in the Kahlert School of Computing, Chemistry, Mechanical Engineering, Sociology and Criminology, and other departments in Utah.

I have built successful collaborations with researchers in national laboratories including Pacific Northwest National Laboratory (PNNL), Air Force Research Laboratory (AFRL), and Oak Ridge National Laboratory (ORNL). Several of my PhD students have been funded for long term (9–12 months or more) internships at these institutions for working on research projects that I also am actively involved in. I have a current consulting collaboration with Mitsubishi Electric Research Labs (MERL). At Utah, I will try to expand such collaborations and actively involve my students and colleagues in them.

I enjoy mentoring students a lot—I have worked with 23 PhD students (17 graduated, 6 current), 10 MS students, and 16 undergraduate students. I have also maintained an excellent record as a teacher at WSU, as testified by the uniformly high course evaluation scores (see my CV) as well as multiple excellence in teaching awards at the College and University levels. Teaching at Utah may offer new challenges, which I am confident of overcoming effectively.

I also bring in valuable leadership experience building the Mathematics and Statistics graduate program up from scratch at our Vancouver, WA, campus (most departments in WSU operate across multiple campus locations). I moved to the Vancouver campus in 2014 and established the part of our MS+PhD program in that location in coordination with the department that is mostly based in our Pullman campus. I served as Associate Chair (Program Leader) in Vancouver from 2015–2019 to effect the establishment of this program. I am also the current Director of Graduate Programs for the whole department (since August 2026). Apart from the standard duties that this role entails, I am leading a Professional Development Seminar for our senior and graduating PhD students that focuses on upskilling the students to be better prepared for industry and government job searches (in addition to the standard academic job search focused on both research and teaching).

I am truly excited by the possibility of joining you at Utah. Please let me know if I can give any more information. I look forward to hearing from you.

Yours sincerely,



Bala Krishnamoorthy